Course 02323/02402 Introduction to Statistics

Week 7: Simulation-based statistics

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Agenda

1. For appendix: Python commands for t-test
2. Introduction to Simulation
   - Example: Area of Plates
3. Propagation of error

Overview

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Python commands for t-test

t-test with one sample:

- `stats.ttest_1samp(data, pop_mean = mu0)`

Welch t-test with two samples:

- `stats.ttest_ind(dataA, dataB, equal_var = False)`

Pooled t-test with two samples:

- `stats.ttest_ind(dataA, dataB, equal_var = True)`

Paired t-test with two samples:

- `stats.ttest_rel(dataA, dataB)`
- `stats.ttest_1samp(diff_data, pop_mean = 0)`



(x5)

In addition, the confidence interval can be calculated with (here showing 95% CI):

- `stats.ttest_****(...).confidence_interval(0.95)`

Python commands for strength calculations

One-sample test:

- `smp.TTestPower().solve_power(...)`

$$n = \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2$$

Two-sample test:

- `smp.TTestIndPower().solve_power(...)`
- also provide:
  `nobs1` $= n_1$
  and
  `ratio` $= n_2/n_1 (= 1/k)$

$$n = (k+1) \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_1 - \mu_2} \right)^2$$

$k = n_1/n_2$

(NOTE: this formula is not in the book)

NOTE: The Python commands do not calculate exactly the same as the formulas!

Python commands for power calculations

One-sample test (excerpt from documentation):

```
TTestPower.solve_power(
    effect_size=None,
    nobs=None,
    alpha=None,
    power=None,
    alternative='two-sided'
) ¶                                                    [source]
```

solve for any one parameter of the power of a one sample t-test

for the one sample t-test the keywords are:

effect_size, nobs, alpha, power

Exactly one needs to be `None`, all others need numeric values.

Python commands for power calculations

One-sample test (excerpt from documentation):

```
TTestIndPower.solve_power(
    effect_size=None,
    nobs1=None,
    alpha=None,
    power=None,
    ratio=1.0,
    alternative='two-sided'
) ¶                                                    [source]
```

solve for any one parameter of the power of a two sample t-test

for t-test the keywords are:

effect_size, nobs1, alpha, power, ratio

exactly one needs to be `None`, all others need numeric values



(x1)

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Motivation

We have already used simulation:

- To simulate dice rolls
- To understand the motivation behind the CLT (Central Limit Theorem)
- To test various scenarios related to samples

In the past few weeks, we have worked with hypothesis testing for mean(s).

With simulation, we can investigate many other quantities — including random quantities that follow more complex distributions.

What is simulation actually?

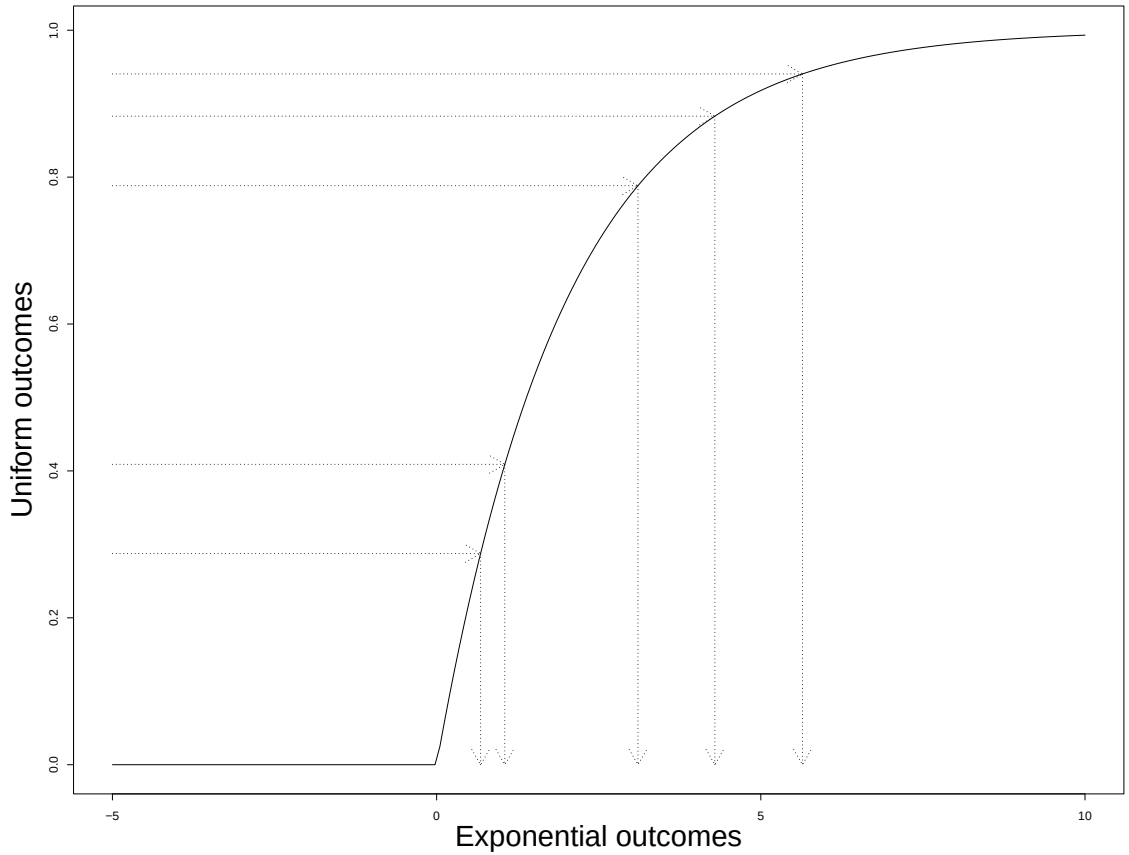
- (Pseudo)-random numbers generated by a computer.
- A *random number generator* is an algorithm that can generate pseudo-random numbers.
- The algorithm requires a so-called *seed* (often the current time is used if another seed is not specified).
- You can simulate from (almost) any distribution by inverting the distribution function F and inserting uniformly distributed numbers:

Theorem 2.51: All distributions can be "simulated" from the uniform distribution

If $U \sim \text{Uniform}(0, 1)$ and F is the distribution function for a given probability distribution, then $F^{-1}(U)$ will follow the distribution given by F.

Example: Exponential Distribution with $\lambda = 0.5$

$$F(x) = \int_0^x f(t)dt = 1 - e^{-0.5x}$$



In Practice in Python

Many distributions are available via the `scipy.stats` package, and you can simulate random numbers using the `.rvs()` function (stands for *random variates*):

`scipy.stats.binom.rvs()`	Binomial distribution
`scipy.stats.poisson.rvs()`	Poisson distribution
`scipy.stats.hypergeom.rvs()`	Hypergeometric distribution
`scipy.stats.norm.rvs()`	Normal distribution
`scipy.stats.lognorm.rvs()`	Log-normal distribution
`scipy.stats.expon.rvs()`	Exponential distribution
`scipy.stats.uniform.rvs()`	Uniform distribution
`scipy.stats.t.rvs()`	t-distribution
`scipy.stats.chi2.rvs()`	χ^2-distribution
`scipy.stats.f.rvs()`	F-distribution

Example: Area of Plates

A company produces rectangular plates.

The length of the plates (in meters), X, is assumed to be described by a normal distribution $N(2, 0.01^2)$, while the width of the plates (in meters), Y, is assumed to be described by a normal distribution $N(3, 0.02^2)$. It can be assumed that the lengths and widths of the plates are independent.

We are interested in the area, A, which is given by $A = XY$.

- What is the mean area?
- What is the variation in the area from plate to plate?
- How often do such plates have an area that deviates more than 0.1 m^2 from the specified 6 m^2?
- Probabilities for other events.
- In general: What is the distribution of the random variable A?

Example: Area of Plates – Solution by Simulation

Kahoot!

(x1)

- Go to today's Python notebook in VS Code
  - "Example: Area of Plates"



Visual Studio Code

(+ Kahoot x1)

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Propagation of error

We wish to find:

$$\sigma^2_{f(X_1,\ldots,X_n)} = \mathrm{Var}(f(X_1,\ldots,X_n))$$

Linear combination of independent variables:

$\sigma^2_{f(X_1,\ldots,X_n)} = \sum_{i=1}^n a_i^2\sigma_i^2$ when $f(X_1,\ldots,X_n) = \sum_{i=1}^n a_i X_i$ (with independence)

Method 4.3 The non-linear approximative error propagation rule

If $X_1,\ldots,X_n$ are independent random variables with variances $\sigma_1^2,\ldots,\sigma_n^2$ and f is a (potentially non-linear) function of n variables, then the variance of the f-transformed variables can be approximated linearly by

$$\sigma^2_{f(X_1,\ldots,X_n)} = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_i^2, \qquad (4\text{-}2)$$

where $\frac{\partial f}{\partial x_i}$ is the partial derivative of f with respect to the i'th variable

Example: Area of Plates (continued)

In the first part of the lecture, we used simulation to investigate the variance of the area of plates.

We can also calculate the variance of $A = XY$ using the error propagation law.

For this calculation, we need the following values:

$$\mathbf{E}[X] = 2$$
$$\mathbf{E}[Y] = 3$$
$$\mathbf{V}[X] = 0.01^2$$
$$\mathbf{V}[Y] = 0.02^2$$

Example: Area of Plates (continued)

The variances are:

$$\sigma_1^2 = \mathbf{V}[X] = 0.01^2 \text{ and } \sigma_2^2 = \mathbf{V}[Y] = 0.02^2.$$

Example: Area of Plates (continued)

The variances are:

$$\sigma_1^2 = \mathbf{V}[X] = 0.01^2 \text{ and } \sigma_2^2 = \mathbf{V}[Y] = 0.02^2.$$

The function and its partial derivatives are:

$$f(x,y) = xy, \ \frac{\partial f}{\partial x} = y, \ \frac{\partial f}{\partial y} = x.$$

Example: Area of Plates (continued)

The variances are:

$$\sigma_1^2 = \mathbf{V}[X] = 0.01^2 \text{ and } \sigma_2^2 = \mathbf{V}[Y] = 0.02^2.$$

The function and its partial derivatives are:

$$f(x,y) = xy, \ \frac{\partial f}{\partial x} = y, \ \frac{\partial f}{\partial y} = x.$$

So the result becomes:

$$\begin{aligned} \mathbf{V}[A] = \sigma_A^2 &\approx \left(\frac{\partial f}{\partial x}\right)^2 \sigma_1^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_2^2 \\ &= y^2\sigma_1^2 + x^2\sigma_2^2 \\ &= 3.00^2 \cdot 0.01^2 + 2.00^2 \cdot 0.02^2 \\ &= 0.0025. \end{aligned}$$

Non-linear Error Propagation – by Simulation

Method 4.4: Non-Linear Error Propagation by Simulation

Assume we have (actual) measurements $x_1, \ldots, x_n$ with known/assumed (estimated) variances $\sigma_1^2, \ldots, \sigma_n^2$.

1. Simulate k outcomes of all n measurements from the assumed distributions, for example $X_i^{(j)} \sim N(x_i, \sigma_i^2)$, $j = 1, \ldots, k$, $i = 1, \ldots, n$.
2. Calculate the standard deviation directly as the observed standard deviation of the k simulated values of $f(X_1^{(j)}, \ldots, X_n^{(j)})$:

$$s^{\text{sim}}_{f(X_1,\ldots,X_n)} = \sqrt{\frac{1}{k-1}\sum_{i=1}^{k}(f_j - \bar{f})^2}$$

where

$$f_j = f(X_1^{(j)}, \ldots, X_n^{(j)}).$$

Example: Area of Plates (continued)

We can derive the variance of $A = XY$ theoretically (from ex. 4.5):

following fundamental relationship

$$\mathrm{V}(X) = \mathrm{E}(X - \mathrm{E}(X))^2 = \mathrm{E}(X^2) - \mathrm{E}(X)^2. \tag{4-5}$$

So, one can actually deduce the variance of A theoretically, it is only necessary to know in addition that for independent random variables: $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$ (which by the way then also tells us that $\mathrm{E}(A) = \mathrm{E}(X)\,\mathrm{E}(Y) = 6$)

$$\begin{aligned}
\mathrm{V}(XY) &= \mathrm{E}\left[(XY)^2\right] - \mathrm{E}(XY)^2 \\
&= \mathrm{E}(X^2)\,\mathrm{E}(Y^2) - \mathrm{E}(X)^2\,\mathrm{E}(Y)^2 \\
&= \left[\mathrm{V}(X) + \mathrm{E}(X)^2\right]\left[\mathrm{V}(Y) + \mathrm{E}(Y)^2\right] - \mathrm{E}(X)^2\,\mathrm{E}(Y)^2 \\
&= \mathrm{V}(X)\,\mathrm{V}(Y) + \mathrm{V}(X)\,\mathrm{E}(Y)^2 + \mathrm{V}(Y)\,\mathrm{E}(X)^2 \\
&= 0.01^2 \cdot 0.02^2 + 0.01^2 \cdot 3^2 + 0.02^2 \cdot 2^2 \\
&= 0.00000004 + 0.0009 + 0.0016 \\
&= 0.00250004.
\end{aligned}$$

Note, how the approximate error propagation rule actually corresponds to the two latter terms in the correct variance, while the first term – the product of the two variances is ignored. Fortunately, this term is the smallest of the three in this case. It does not always have to be like that. If you want to learn how to make a theoretical derivation of the density function for $A = XY$ then take a course in probability calculation.

Example: Area of Plates (continued)

Three Different Approaches:

1. Simulation
2. The approximate *Propagation of error*
3. Theoretical derivation

Example: Area of Plates (continued)

Three Different Approaches:

1. Simulation
2. The approximate *Propagation of error*
3. Theoretical derivation

The simulation approach has some important advantages:

1. Easy way to compute other quantities than SEM.
2. Easy way to use other distributions than the normal distribution.
3. Does not depend on a linear approximation of the underlying non-linear function (unlike the error propagation law).

Exam Question

Question (20) from the May 2024 exam:

Exercise XI

In a laboratory, one wants to dilute a stock solution A of concentration C_A to obtain a solution B of concentration C_B. The following rule is applied:

$$C_B = \frac{C_A V_A}{V_B}$$

Here, V_A and V_B represent the volumes of the initial stock solution A and the resulting solution B, respectively. C_A, C_B, V_A and V_B are all random variables.

Question XI.1 (20)

Which of the following formulas can be utilized to approximate the standard deviation of solution B's concentration (σ_{C_B}) via the non-linear error propagation rule?

Exam Question

$$C_B = \frac{C_A V_A}{V_B}$$

1 ☐ $\sigma_{C_B} = \sqrt{\frac{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2}{\left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}}$

2 ☐ $\sigma_{C_B} = \left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2$

3 ☐ $\sigma_{C_B} = \sqrt{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}$

4 ☐ $\sigma_{C_B} = \frac{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2}{\left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}$

5 ☐ $\sigma_{C_B} = \left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right) + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right) + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)$

Kahoot!

(x1)

Exam Question

What would one actually substitute into the formula? $C_B = \frac{C_A V_A}{V_B}$

1 ☐ $\sigma_{C_B} = \sqrt{\frac{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2}{\left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}}$

2 ☐ $\sigma_{C_B} = \left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2$

3 ☐ $\sigma_{C_B} = \sqrt{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}$

4 ☐ $\sigma_{C_B} = \frac{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2}{\left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}$

5 ☐ $\sigma_{C_B} = \left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right) + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right) + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)$

Solving the exam question

The variance can be approximated linearly by (4-2)

$$\sigma^2_{f(X_1,\ldots,X_n)} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial X_i}\right)^2 \sigma_i^2,$$

where $\frac{\partial f}{\partial X_i}$ is the partial derivative of f with respect to the i'th variable.
Here $f = C_B$ and the inputs are C_A, V_A, V_B. So, the partial derivatives are:

$$\frac{\partial C_B}{\partial C_A}, \frac{\partial C_B}{\partial V_A}, \frac{\partial C_B}{\partial V_B}$$

$$\sigma_{C_B} = \sqrt{\left(\frac{\partial C_B}{\partial C_A}\sigma_{C_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_A}\sigma_{V_A}\right)^2 + \left(\frac{\partial C_B}{\partial V_B}\sigma_{V_B}\right)^2}.$$

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